

Cambridge International AS & A Level

CANDIDATE
NAME

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CENTRE
NUMBER

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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

May/June 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

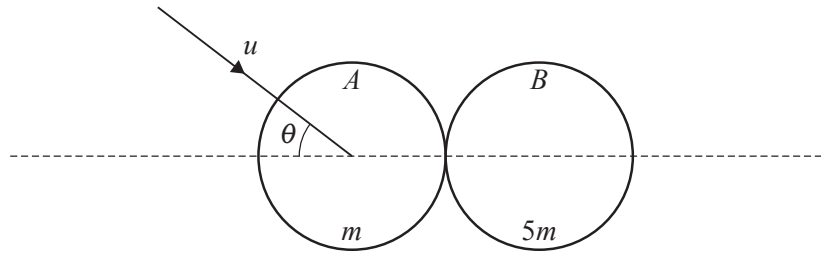
INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.



Two smooth uniform spheres A and B of equal radii have masses m and $5m$ respectively. Sphere A is moving on a smooth horizontal surface with speed u when it collides with sphere B which is at rest on the surface. Immediately before the collision, A 's direction of motion makes an angle of θ with the line of centres. After the collision, the kinetic energies of A and B are equal. The coefficient of restitution between the spheres is $\frac{1}{2}$.

Find the value of $\tan \theta$.

[6]

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dashed lines spaced evenly apart, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings present.

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- This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 28 evenly spaced, horizontal dotted lines extending across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

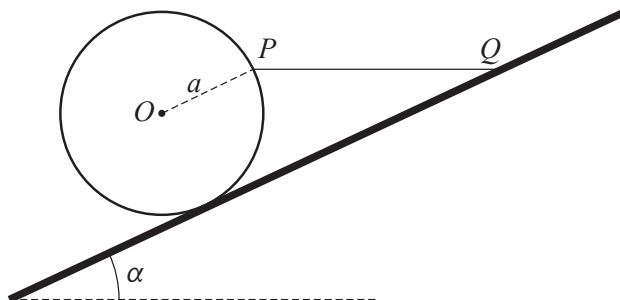
The particle P is pulled down vertically and released from rest at a distance $\frac{8}{5}a$ below M .

- (b)** Find, in terms of a and g , the speed of P as it passes through M in the subsequent motion. [4]

[illegible]

[5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



A ring of weight W , with radius a and centre O , is at rest on a rough surface that is inclined to the horizontal at an angle α where $\tan \alpha = \frac{1}{2}$. The plane of the ring is perpendicular to the inclined surface and parallel to a line of greatest slope of the surface. The point P on the circumference of the ring is such that OP is parallel to the surface.

A light inextensible string is attached to P and to the point Q , which is on the surface, such that PQ is horizontal (see diagram). The points O , P and Q are in the same vertical plane. The system is in limiting equilibrium and the coefficient of friction between the ring and the surface is μ .

- (a) Find, in terms of W , the tension in the string PQ . [4]

[illegible]

[Turn over

(a) Find the tension in the string, in terms of m and g . [4]

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 20 evenly spaced, horizontal dotted lines running across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

(b) Find the value of k .

[3]

[illegible]

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- This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

(b) Find an expression for x in terms of t .

[4]

[illegible]

[illegible]

It is given that $\cos \theta = \frac{5}{6}$.

- (b)** Find, in terms of a and g , an expression for the vertical component of the velocity of P just before it hits the horizontal plane to which the sphere is fixed. [3]

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice. There are no margins, text, or other markings on the page.

- (c) Find an expression for the time taken by P to fall from Q to the plane. Give your answer in the form $k\sqrt{\frac{a}{g}}$, stating the value of k correct to 3 significant figures. [2]

This image shows a full page of white paper with ten evenly spaced horizontal dashed lines, typical of primary school handwriting practice paper. The lines extend across the entire width of the page.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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